SmartWealth: Goal-Based Investment Strategy for a Digital Wealth Platform

Inter-IIT Tech Meet 15.0 – Practice Problem Statement

1 Business Context

NiveshNow is a fictional Indian digital wealth platform that wants to help users connect their investments with their financial goals. Many investors struggle to choose products that fit their goal amount, time horizon, risk tolerance, liquidity needs, and ability to contribute regularly. Some may take excessive risk for short-term expenses, while others may leave too much money idle for long-term goals.

Your team must build a prototype called **SmartWealth** that converts an investor profile and a supplied catalogue of fictional investment products into a suitable, cost-aware, and explainable investment plan. The system should also estimate whether the goal appears feasible under stated assumptions and show how the plan changes when circumstances change.

This is an applied FinTech and wealth-technology product-engineering challenge. The goal is not to predict market prices, generate trading signals, or maximise returns at any cost. A transparent rule-based approach is acceptable; constrained optimisation may be used if it is implemented and explained clearly.

2 Challenge Objectives

Your solution should address the following objectives.

1. **Investor profiling and suitability:** Use the goal amount, time horizon, existing investment, monthly contribution (SIP), risk tolerance, emergency-fund coverage, liquidity needs, investment experience, and any optional preference. Detect missing, invalid, or contradictory inputs and handle them explicitly.
2. **Portfolio allocation:** Allocate the available investment amount across the supplied fictional products. Recommendations must account for suitability, horizon, risk, liquidity, and cost; selecting products only because they have the highest assumed return is not sufficient.
3. **Goal feasibility:** Estimate a range of possible future values, compare them with the target, and report the estimated surplus or shortfall in INR. Explain practical adjustments where a goal appears infeasible. Do not present projections as guaranteed outcomes.
4. **Cost and liquidity trade-offs:** Report the weighted annual expense ratio and estimated annual product cost. Consider liquidity and settlement time, minimum investment, and lock-in periods.
5. **Explainability:** Explain why recommended products were included, which investor constraints they serve, the main downside or trade-off, and why a plausible alternative was limited or excluded.
6. **What-if analysis:** Demonstrate how recommendations and projections respond to at least two changes, such as a 15% market shock, a 20% reduction in monthly SIP, a shorter goal horizon, or a change in risk tolerance.

3 Input Files Supplied

The following three CSV files are supplied with this problem statement. Use these files as the public development dataset. Participants are not required to obtain external datasets, connect to live financial APIs, or use a brokerage account.

File	Contents and intended use
`investor_profiles.csv`	Twelve synthetic investor personas with goal amount, horizon, existing investment, monthly SIP, risk tolerance score, emergency-fund coverage, liquidity need, experience, and optional preference.
`investment_products.csv`	Eight fictional investment-product categories, including asset class, risk score, assumed annual return, volatility proxy, annual expense ratio, liquidity days, minimum investment, and lock-in.
`synthetic_monthly_returns.csv`	Thirty-six months of synthetic monthly returns for the supplied products. This data may be used for consistency checks or simulations; it is not historical market data and must not be presented as real-world performance.

The supplied data is intended to help you develop and test your strategy. Your solution should operate on investor inputs, rather than being hard-coded only for the supplied investor IDs. During evaluation, the organisers may run submitted solutions on additional investor profiles and altered scenarios that are not included in the public files. The exact hidden cases will not be disclosed in advance. Your system should therefore generalise to valid, unseen inputs while continuing to respect the stated constraints.

3.1 Minimum Data Validation

At a minimum, your system must:

- Ensure allocation weights are non-negative and sum to 100% within a tolerance of 0.1 percentage points.
- Check that required fields are present and numeric fields are valid and finite.
- Confirm that every referenced product ID exists in the supplied catalogue.
- Prevent invalid or non-finite projection outputs and clearly handle malformed inputs.
- Return a clear warning, exception, or safer fallback when the inputs cannot support a normal recommendation.

4 Suitability and Risk Guardrails

Your system must implement the following minimum guardrails. You may introduce stricter safeguards, but you must document them and apply them consistently.

1. **Diversification:** No single product may receive more than 40% of the portfolio. Use at least two product categories unless there is a clear, profile-specific reason not to; explain any exception.
2. **Horizon-based allocation limits:** Apply the following limits to the combined allocation in the specified product risk-score groups.

Goal horizon	Risk score 4–5	Risk score 1–2
Up to 36 months	Maximum 10% combined	Minimum 70% combined
37–84 months	Maximum 30% combined	Minimum 45% combined
More than 84 months	Maximum 55% combined	Minimum 25% combined

3. **Risk-tolerance preference:** If the investor's risk tolerance requires a stricter limit than the horizon-based limit, apply the stricter limit. The weighted portfolio risk score must not exceed the investor's risk-tolerance score by more than 0.25. If the requirements conflict or no feasible allocation is found, return a clear exception and a safer fallback rather than silently violating constraints.
4. **Liquidity and lock-in:** Report whether the allocation available within the required time covers the stated liquidity need. Do not allocate funds to a product whose lock-in extends beyond the goal date.
5. **Emergency buffer:** If emergency-fund coverage is below three months, display a prominent warning and demonstrate a conservative, liquid-buffer option before allocating all available capital to the long-term goal.
6. **Costs:** Report the weighted annual expense ratio and estimated annual product cost based on the allocated balance. Do not compromise suitability or liquidity merely to reduce costs.
7. **No false certainty:** Use only the supplied assumptions and clearly label outputs as educational estimates. Do not imply guaranteed returns or provide claims of actual investment performance.

5 Goal Projection Method

Use a documented and consistent method to estimate the future value of an initial investment and monthly contributions. A simple planning approximation is acceptable.

Let P be the initial amount invested, S the monthly SIP, r the assumed annual portfolio return expressed as a decimal, and n the number of months. The following approximation may be used:

$$FV(P, S, r, n) = P\left(1 + \frac{r}{12}\right)^n + S\left[\frac{\left(1 + \frac{r}{12}\right)^n - 1}{r/12}\right].$$

When $r = 0$, use $FV = P + Sn$. For portfolio weights w_i and product assumed annual returns r_i, the weighted assumed annual return is $r_p = \sum_i w_i r_i$. For a basic scenario range, use $r_p - 3$ percentage points, r_p, and $r_p + 3$ percentage points, with a sensible floor or cap if used and a clear explanation of the choice. State any additional assumptions, including the treatment of contributions, fees, and timing. This formula is a planning approximation, not a forecast guarantee.

6 Required User Experience

The reviewer must be able to select an investor persona and inspect the recommendation. The prototype should display:

- Product-level allocation as both percentages and INR amounts, with an asset-class breakdown.
- Weighted portfolio risk score, weighted annual expense ratio, and estimated annual product cost.
- Goal projection range, target amount, and estimated shortfall or surplus.

- At least two profile-specific reasons supporting the recommendation.
- Relevant warnings, including emergency-buffer concerns, liquidity or lock-in mismatch, short horizon, cost considerations, and goal infeasibility.
- Updated outputs after changing at least one meaningful input, such as monthly SIP, goal horizon, or risk tolerance.

A Streamlit application, React application, or well-structured command-line prototype with a clear demonstration is acceptable. Interface polish does not compensate for incorrect calculations or unsuitable recommendations. Do not use live prices or real personal financial data.

7 Submission Package

Submit the following items:

1. **Runnable prototype:** Include source code, dependencies, setup instructions, and a straightforward launch command.
2. **Recommendation outputs:** Provide a CSV with one row per client/product allocation and a summary CSV containing goal projections, risk fit, costs, liquidity status, and warnings.
3. **Technical report:** Maximum five pages, covering architecture, recommendation logic, assumptions, trade-offs, failure cases, and possible improvements.
4. **Demonstration:** A maximum five-minute demonstration covering a conservative short-horizon investor, a long-horizon investor, and an edge case or infeasible goal.
5. **Automated tests:** Include at least five tests covering allocation sums, non-negative weights, lock-in/horizon rules, risk limits, and malformed investor profiles.

Generalisation requirement: The submitted system must accept investor data through a reusable function, API, or command-line interface rather than relying on hard-coded outputs for the public personas. The organisers may use a standardised input/output contract to test submissions on unseen profiles and scenario changes. Ensure that your recommendation logic can be rerun with different valid inputs without manual code edits. Teams may use different valid allocation strategies; a single predetermined portfolio is not required.

8 Evaluation Rubric

Submissions will be evaluated out of 100 points using the following rubric.

Criterion	**Weight**
Suitability and guardrail compliance, including risk, horizon, liquidity, and emergency-buffer handling	25
Goal planning, projection correctness, and shortfall analysis	20
Product and workflow quality, including ability to handle unseen profiles and scenario changes	15
Explainability, transparency, and user trust	15
Engineering correctness, validation, and automated tests	10
Cost and liquidity trade-off reasoning	10
Robustness and communication	5
Total	**100**

Evaluation will consider three complementary aspects: (i) compliance with the stated rules, (ii) correctness of independently checkable calculations, and (iii) the quality and consistency of decisions on unseen profiles or changed scenarios. Teams are not expected to produce identical allocations. Different approaches may receive full credit when they are financially consistent, respect the guardrails, and justify their trade-offs. The synthetic returns are not a basis for ranking teams by real-world performance, Sharpe ratio, or CAGR.

9 Important Notes

- All investor profiles, products, and return observations supplied for this challenge are synthetic.
- Do not describe assumed returns or simulated projections as guaranteed outcomes or actual market performance.
- Make assumptions explicit and provide sensible behaviour for infeasible goals, conflicting constraints, and invalid inputs.
- The purpose is to build a responsible and explainable goal-based planning prototype, not a trading system or a substitute for professional financial advice.